

SmartInterestX — Loan & Interest Management Mobile Application

Project Overview

SmartInterestX is a fintech-style mobile application designed to help users **track, calculate, and manage interest-based transactions** between borrowers and lenders.

The app automates:

- Loan & interest calculations
- Due date tracking
- Payment proof management
- Push notifications for reminders
- Analytics dashboard for interest earned/paid

This project simulates real-world finance tools used by:

- Small businesses
- Individual lenders
- Finance consultants
- Micro-lending agencies

Students will build a **production-style mobile application** with:

- Clean UI/UX (as per provided PDF)
- Local & cloud data management
- Notifications & reminders



- Secure backup & export options

Project Objective

To design and develop a **loan & interest tracker mobile app** that allows users to:

- Add **borrower/lender entries**
- Automatically compute interest (monthly/yearly)
- Get **reminders for due dates**
- View **analytics on total interest earned/paid**
- Store **payment proof**
- Backup & export transaction data

Core Features (Must-Have)

1. Loan/Transaction Entry Module

- Add new transaction:
 - Amount
 - Type: **Given** (you lent) / **Taken** (you borrowed)
 - Interest rate (%)
 - Start date
 - Due date (optional)
- Tag as **Borrower** or **Lender**
- Add notes/description



2. Borrower/Lender Database

- Save contact details:
 - Name
 - Mobile number
 - Email (optional)
- Link multiple transactions to the same person
- Quick search & filter by name or type

3. Interest Calculation Engine

- Support:
 - Simple interest (monthly / yearly)
- Auto-calculation of:
 - Interest amount
 - Total amount payable/receivable
- Option to view:
 - Interest till today
 - Interest till due date

4. Payments & Proof Management

- Mark a transaction as **partially paid** or **fully settled**
- Record payment details:
 - Payment date



- Mode: UPI / Bank Transfer / Cash / Other
- Upload or attach:
 - Screenshot/receipt as **proof**
- Maintain payment history per transaction

5. Notifications & Reminders

- Push notifications for:
 - Upcoming due dates
 - Overdue payments
 - Monthly summary of interest earned/paid
- Custom reminder settings (1 day/3 days/7 days before due date)

6. Dashboard & Analytics

- Overview cards:
 - Total interest **earned**
 - Total interest **paid**
 - Total **outstanding receivables & payables**
- Filters:
 - By month
 - By year
 - By contact



- Charts for:
 - Given vs Taken
 - Monthly interest flow

7. Data Export & Backup

- Export transactions to:
 - CSV / Excel
- Backup options:
 - Local backup (on device)
 - Cloud backup (Firebase / Google Drive API – recommended)

Suggested Technical Framework

You can adapt based on your stack, but here is a solid, industry-style recommendation:

Frontend

- **Flutter** (recommended)
 - Single codebase for Android & iOS
 - Matches your existing app development course

State Management

- **Provider** or **Riverpod**
- Alternative: **Bloc / GetX** (if already taught)



Local Database

Any one of:

- **sqflite** — SQL-like local DB
- **hive** — Lightweight NoSQL storage

Use it to store:

- Borrower/lender data
- Transactions
- Interest calculation metadata

Cloud Backend (Optional but Recommended)

- **Firebase Firestore** for cloud sync & backup
- **Firebase Storage** for payment proof images
- **Firebase Authentication** (optional for login)

Notifications

- **Local Notifications:**
 - **flutter_local_notifications** – scheduled reminders
- **Push Notifications (Cloud):**
 - **firebase_messaging**

File Handling & Export



- `csv` / `excel` packages for export
- `path_provider` for storage paths
- `share_plus` to share exports

Security & Backup

- Basic:
 - App-level PIN or biometric lock (using `local_auth`)
- Backup:
 - Export data as encrypted JSON/CSV
 - Cloud sync using Firestore / Drive APIs

Core App Modules (Technical Breakdown)

1. **Auth & Onboarding (Optional, but Good Practice)**
 - Splash screen
 - Intro/onboarding screens
 - (Optional) Simple login (e.g., PIN or Google Sign-In)
2. **Contact Management Module**
 - CRUD for borrowers/lenders
 - Link contacts to transactions
3. **Transaction & Interest Module**
 - Create transaction form



- Edit & update entries
- Interest calculation logic (function/service layer)

4. Reminders & Notifications Module

- Schedule notifications for each due transaction
- Handle click actions (open transaction detail screen)

5. Dashboard & Analytics Module

- Aggregation functions for totals
- Chart rendering (using `fl_chart`)

6. Export & Backup Module

- Serialize data
- Export as CSV
- Save to local storage / share

Final Project Deliverables (Student Submission)

Students must submit:

- 1. Complete Source Code (Flutter Project)**
- 2. APK Build (Debug/Release)**
- 3. Database Schema / ER Diagram**
- 4. Feature-wise Screenshots (All Key Screens)**
- 5. Short User Flow Diagram (From Home → Add → View → Pay → Analytics)**
- 6. Technical Documentation, including:**



- Tech stack used
- Data model overview
- Interest calculation logic
- Notification handling
- Backup & export explanation

7. Demo Video (3–5 minutes)

- Show add transaction
- Show interest calculation
- Show reminders & notifications
- Show dashboard & export

Learning Outcomes

By completing this project, students will:

- Build a **real-world fintech-style app**
- Practice **Flutter UI + Firebase + Local DB** integration
- Implement **business logic** (interest calculation)
- Work with **notifications & reminders**
- Design a **clean, user-focused UI** based on a provided PDF
- Handle **app data architecture, backup, and export flows**
- Create a **portfolio-ready project** aligned with 2025 app development standards

